Let us show that:

$$T(X_s) = \operatorname{diag}(\gamma^* \mathbf{1}_{n_t})^{-1}(\gamma^* X_t)$$

where:

- X_t: matrix of size $n_t \times d$ whose each row is a target point
- $T(X_s)$: matrix of size $n_s \times d$ whose each row is a transformed source point

$$T(X_s)_i = T(x_s^i) = \frac{\sum_{j=1}^{n_t} \gamma^*(i,j)\, x_j^t}{\sum_{j=1}^{n_t} \gamma^*(i,j)}$$

1 Proof of the numerator

$$\sum_j \gamma^*(i,j)\, x_j^t = (\gamma^* X_t)_i$$

Indeed:

$$(\gamma^* X_t)_{i,k} = \sum_j \gamma^*(i,j)\, X_t(j,k) = \sum_j \gamma^*(i,j)\, x_{j,k}^t$$

γ^*: matrix of size $n_s \times n_t$
$\mathbf{1}_{n_t}$: column vector of size $n_t \times 1$

$$(\gamma^* \mathbf{1}_{n_t})_{i,1} = \sum_{j=1}^{n_t} \gamma^*(i,j) \cdot 1 = \sum_{j=1}^{n_t} \gamma^*(i,j) = \text{denominator index } i$$

We define:

$$v = \gamma^* \mathbf{1}_{n_t} \quad \text{vector of size } n_s$$

$$v_i = \text{denom. } i$$

$$\operatorname{diag}(v) = \operatorname{diag}(\gamma^* \mathbf{1}_{n_t}) = \begin{pmatrix} v_1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & v_{n_s} \end{pmatrix}$$

$$\operatorname{diag}(v)^{-1} = \begin{pmatrix} \frac{1}{v_1} & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \frac{1}{v_{n_s}} \end{pmatrix}$$

We set

$$\hat{X}_s = \operatorname{diag}(v)^{-1}(\gamma^* X_t) = \operatorname{diag}(\gamma^* \mathbf{1}_{n_t})^{-1}(\gamma^* X_t)$$

$$(\hat{X}_s)_i = \frac{1}{v_i} \cdot (\gamma^* X_t)_i = \frac{1}{v_i} \sum_j \gamma^*(i,j)\, x_j^t = \frac{\sum_j \gamma^*(i,j)\, x_j^t}{\sum_j \gamma^*(i,j)} = T(x_i^s)$$

Therefore we have

$$T(X_s) = \hat{X}_s = \operatorname{diag}(\gamma^* \mathbf{1}_{n_t})^{-1}(\gamma^* X_t)$$